2026 Final Year Projects

Software Requirements Specification

Project: School Digital Platform
Course: COSC300 – Software Development

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1. Introduction

1.1 Purpose

This Software Requirements Specification (SRS) outlines the functional and non-functional requirements of the School Digital Platform. The system is designed to improve communication between schools, students, parents, and teachers while supporting academic monitoring using Artificial Intelligence.

1.2 Document Conventions

When writing this Software Requirements Specification (SRS) document, the following standards and conventions were followed:

- Font: Times New Roman, size 14
- Highlighting: Bold text is used to emphasize important terms and headings
- Numbering: Sections and requirements are numbered for clarity and reference

1.3 Intended Audience and Reading Suggestions

This document is intended for various stakeholders involved in the development and use of the School Digital Platform, including:

- **Developers** – to understand system requirements and implementation details
- **System Designers** – to design the system architecture and database structure
- **Project Managers** – to plan and monitor project progress
- **Teachers and School Administrators** – to understand system functionality
- **Testers** – to verify that the system meets the specified requirements

Readers are encouraged to review the document sequentially, starting from the introduction and proceeding through the system description and requirements sections. Technical users may focus on detailed system requirements, while non-technical stakeholders may focus on general system functionality.

1.4 Scope

The School Digital Platform is a web and mobile-based system that provides centralized access to academic information such as announcements, homework, and student performance. The system includes an Artificial Intelligence component that predicts whether a student is at risk of failing and provides an AI Tutor Expert to support learning.

1.5 References

- Pressman, R. S., & Maxim, B. R. (2020). *Software engineering: A practitioner's approach* (9th ed.). McGraw-Hill.
- Sommerville, I. (2016). *Software engineering* (10th ed.). Pearson.
- Dennis, A., Wixom, B. H., & Roth, R. M. (2015). *Systems analysis and design* (6th ed.). Wiley.
- IEEE. (2018). *IEEE Standard for Software Requirements Specifications*.
- https://www.scribd.com/document/237287232/final-srs

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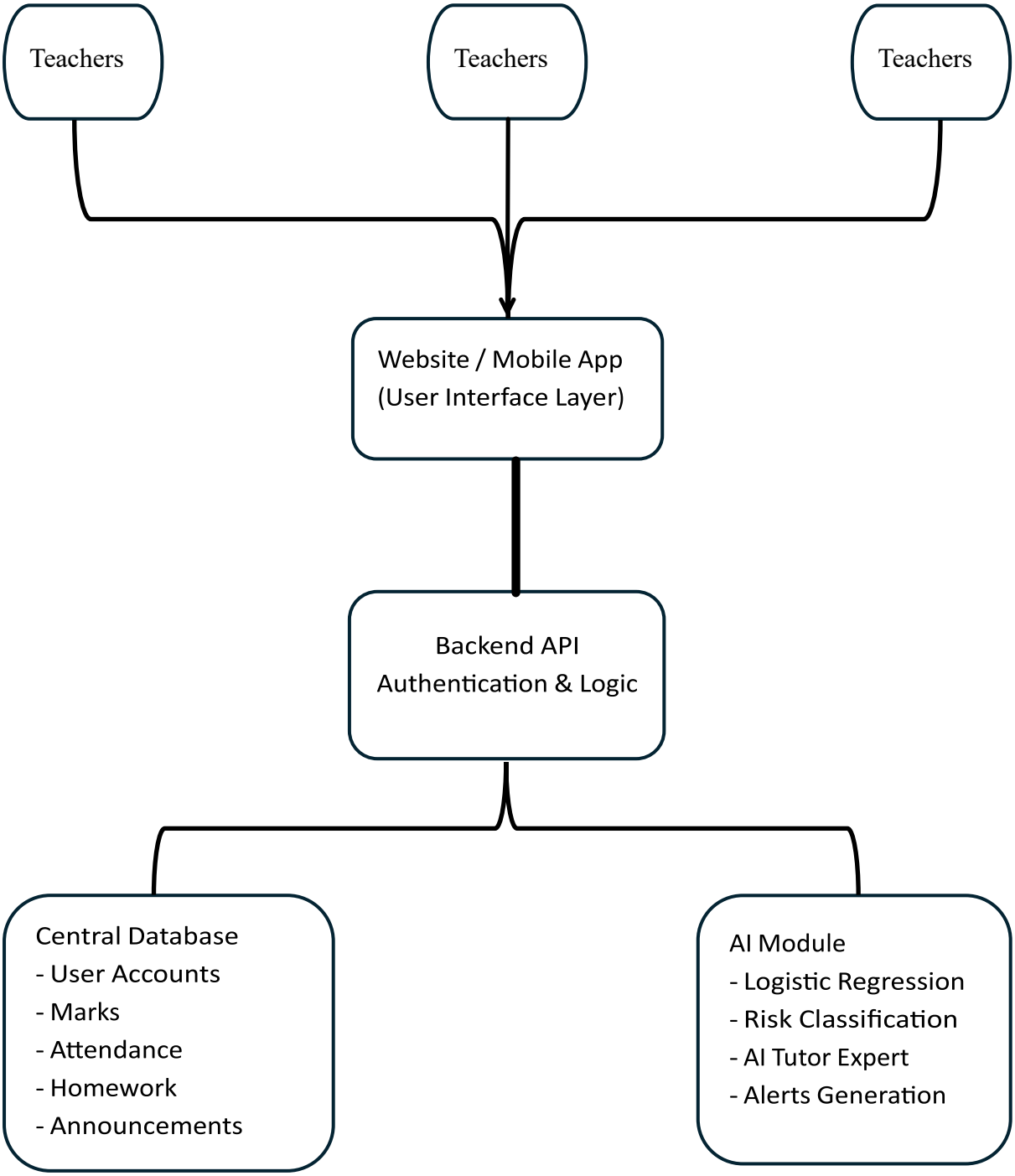
2. Overall Description

2.1 Product Perspective

The system is a centralized digital platform consisting of:

- Website
- Mobile Application
- Backend Server
- Database

- Artificial Intelligence Module



2.2 Product Functions

The system will:

- Allow users to log in securely
- Provide access to announcements and academic materials
- Allow teachers to upload assignments and track performance
- Predict students at risk of failing
- Send alerts to teachers
- Provide AI-based tutoring support

2.3 User Classes and Characteristics

User	Description
Students	Access notes, homework, and AI tutor support
Teachers	Upload content and monitor student performance
Parents	View student progress and announcements
Administrators	Manage users and system operations

2.4 Operating Environment

- Accessible via web browsers (Chrome, Edge) and mobile devices (Android, iOS)
- Hosted on a cloud-based server
- Backend implemented using Node.js or Python
- Database using MySQL or PostgreSQL
- AI model developed using Python (Scikit-learn)
- Requires internet connection

2.5 Design and Implementation Constraints

- Must comply with POPIA for data privacy

- Must support low-end devices
- Must handle unstable internet connectivity
- Must use web and mobile technologies
- AI model must use Logistic Regression

2.6 User Documentation

The system will provide the following documentation:

- User Manual
- Online Help
- Tutorials

2.7 Assumptions and Dependencies

- **Assumptions:**
- Users have access to internet-enabled devices
- Users have basic digital literacy • Student data is available and accurate
- **Dependencies:**
- Cloud hosting services
- Database systems (MySQL/PostgreSQL)
- AI libraries (Scikit-learn)
- Internet connectivity

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3. External Interface Requirements

3.1 User Interfaces

- Login interface
- User dashboards
- Announcement and academic resource interface

- Teacher monitoring interface
- AI Tutor interface
- Notification interface

3.2 Hardware Interfaces

- Smartphones
- Tablets
- Computers

3.3 Software Interfaces

- Database systems (MySQL/PostgreSQL)
- Backend services (Node.js or Python)
- AI module (Python + Scikit-learn)
- Web browsers

3.4 Communication Interfaces

- HTTP/HTTPS protocols
- Email notifications
- Push notifications
- Internet-based communication

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4. Specific Requirements

4.1 System Features

○ Feature 1: Secure User Access

- REQ-1: The system shall authenticate users before access

- REQ-2: The system shall protect user data from unauthorized access

Feature 2: System Performance

- REQ-3: The system shall respond within a few seconds

Feature 3: System Availability

- REQ-4: The system shall be available 24/7

Feature 4: User-Friendly Interface

- REQ-5: The system shall provide an easy-to-use interface

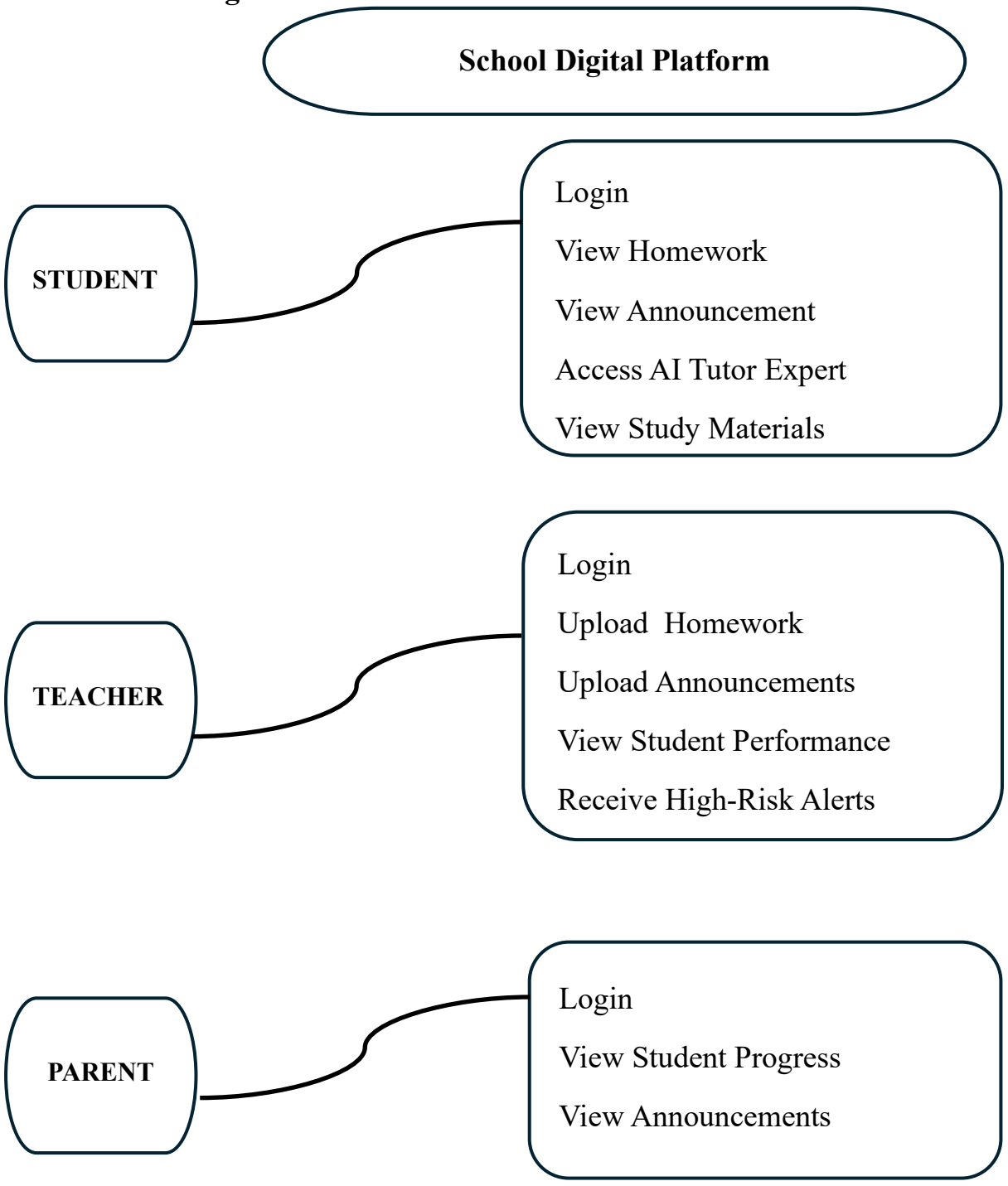
Feature 5: Data Privacy

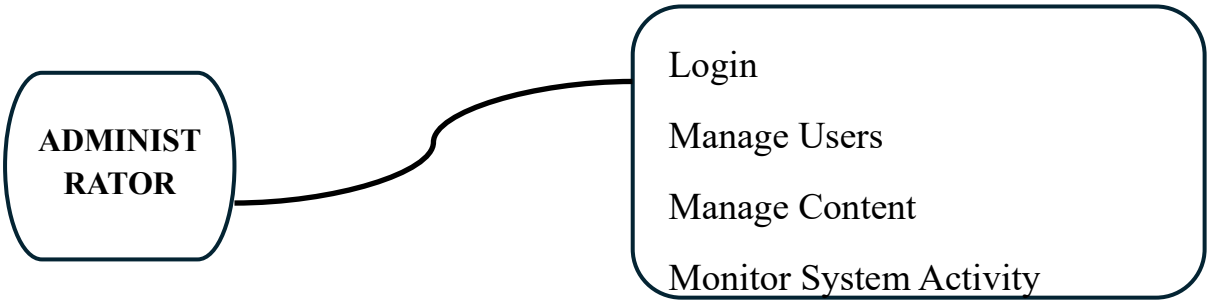
- REQ-6: The system shall comply with POPIA
- REQ-7: The system shall ensure data confidentiality

4.2 Use Cases

- User Login
- View Academic Information
- Monitor Student Performance
- Receive Alerts

4.3 Use Case Diagram





- **Use Case Diagram Description**

The use case diagram illustrates the interaction between users and the School Digital Platform. The main actors include students, teachers, parents, and administrators.

Students can log in, view homework, access announcements, interact with the AI Tutor Expert, and view study materials. Teachers can log in, upload assignments and announcements, monitor student performance, and receive alerts for students identified as high risk. Parents can log in, view student progress, and access announcements. Administrators can manage users, manage system content, and monitor overall system activity.

The diagram shows how each user interacts with the system to perform specific tasks, ensuring effective communication, academic monitoring, and support through Artificial Intelligence.

4.4 Classes / Objects

- **Student**
  - Attributes: StudentID, Name, Marks, Attendance
  - Functions: View results, access materials
- **Teacher**
  - Attributes: TeacherID, Name
  - Functions: Upload content, monitor students
- **AI Model**
  - Attributes: ModelType, Accuracy
  - Functions: Predict performance, classify risk

4.5 Business Rules

- Only authorized users can access the system
- Teachers manage student data
- Students access only their own data
- AI predictions are based on available data

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5. Other Nonfunctional Requirements

5.1 Performance Requirements

- The system shall respond within a few seconds
- The system shall support multiple users

5.2 Safety Requirements

- The system shall prevent data loss
- The system shall provide backup and recovery

5.3 Security Requirements

- Authentication required
- Secure communication (HTTPS)
- Data protection compliance (POPIA)

5.4 Software Quality Attributes

- Usability
- Reliability
- Availability
- Maintainability
- Scalability

6. Artificial Intelligence Requirements

The system will include an Artificial Intelligence module designed to support academic monitoring and learning.

The AI model will analyze:

- Marks
- Attendance
- Homework completion

The system will use a **Logistic Regression algorithm** to predict whether a student will pass or fail.

Students will be classified into:

- Low Risk
- Medium Risk
- High Risk

Teachers will receive alerts for high-risk students.

The system will also include an **AI Tutor Expert** that provides explanations and guidance to students.

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7. Conclusion

This document defines the requirements of the School Digital Platform and provides a clear understanding of system functionality, performance, and AI integration.

Appendix A: Glossary

The glossary defines key terms used in this document.

Term	Definition
SRS	Software Requirements Specification
SDLC	Software Development Life Cycle
AI	Artificial Intelligence
API	Application Programming Interface
Logistic Regression	A machine learning algorithm used to predict outcomes such as pass or fail
Risk Classification	Categorizing students into Low, Medium, or High risk
AI Tutor Expert	An AI-based system that assists students with learning
Backend	The server-side part of the system that processes data
Frontend	The user interface of the system
Database	A system used to store and manage data

Appendix B: Analysis Models

This section includes the models used to analyze and represent the system.

The following models are included in this document:

- **Use Case Diagram** – illustrates how users interact with the system
- **System Architecture Diagram** – shows the structure of the system and how components interact

These models help to provide a clear understanding of system functionality and design, and they support the development of the System Design Document (SDD).

Appendix C: List of References

The following sources were used in the preparation of this Software Requirements Specification:

1. Pressman, R. S., & Maxim, B. R. (2020). *Software engineering: A practitioner's approach* (9th ed.). McGraw-Hill.
2. Sommerville, I. (2016). *Software engineering* (10th ed.). Pearson.
3. Dennis, A., Wixom, B. H., & Roth, R. M. (2015). *Systems analysis and design* (6th ed.). Wiley.
4. IEEE. (2018). *IEEE Standard for Software Requirements Specifications*.